Pimpri Chinchwad Education Trust's

Pimpri Chinchwad College of Engineering (PCCoE), Pune

Department of Computer Engineering — Regional Language



Camera-Only, Continuous Real-Time Marathi Sign Language Recognition:

A Systematic Literature Review of Hybrid 1D-CNN–BiLSTM–Temporal Attention Architectures

Authors (Group No. 19)

Adwait Kamble (123B1D002) | **Yadnesh Patil** (123B1D034) | **Abhay Bormale** (123B1D057) | **Atharva Zope** (123B1D067)

Methodological & Submission Compliance Summary

Review Guideline
PRISMA 2020 [11] & Kitchenham and Charters [12] Systematic Review Protocol

Primary Search Corpus
140 Records across 8 Databases → 15 Primary Studies (S1–S15, Mean QA Score: 9.0/10) + Foundational Literature [1]–[45]

Reference Management
IEEE Citation Style formatted via Mendeley (.docx manuscript)
45 References

Academic Integrity
Turnitin Similarity Index < 10%
AI Writing Detection < 10%

Background, Problem Statement & Motivation



2.1 Background & Paradigm Shift

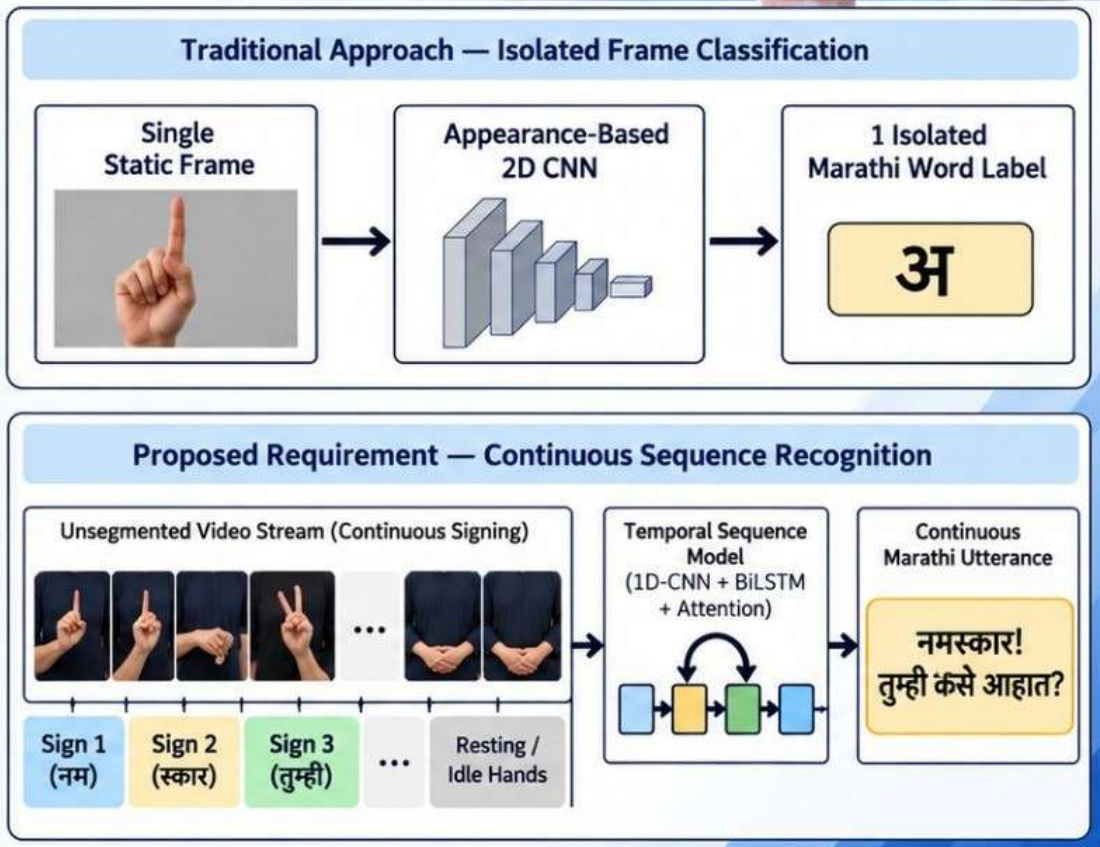
- **Marathi Sign Language (MSL):** Has unique hand gestures, facial expressions and grammar. Certified interpreters are limited, making real-time recognition important for better accessibility.
- **From Isolated to Continuous Signing:** Early systems mainly focused on single-frame (isolated word) recognition, but real conversations are continuous, where signs flow together with co-articulation and natural transitions.

2.2 Key Motivations

- **Accessible Assistive Technology:** Enable MSL recognition using standard laptop/phone cameras without costly hardware.
- **Efficient & Private Processing:** Use lightweight models (e.g., 1D-CNN, BiLSTM with temporal attention) for real-time, on-device inference.
- **Bridge the Resource Gap:** Limited datasets and tools exist for MSL compared to high-resource sign languages, motivating a systematic study.

2.3 Problem Statement

- Existing work mainly handles isolated signs for high-resource languages. For Marathi Sign Language, there is a need for a camera-only, real-time system that can recognize continuous sign sequences accurately while handling co-articulation and motion variations.



Literature Review — Comparative Analysis



No.	Study / Author (Year)	Approach / Method	Dataset / Language	Main Contribution	Limitation / Research Gap
1	**Jiang et al. [6]**	1D-CNN	WLASL (ASL)	• Used 1D-CNN for isolated sign recognition. • Effective spatial feature extraction.	• Not designed for continuous signing.
2	**Pu et al. [11]**	CNN + BiLSTM	RWTH-PHOENIX (ISL)	• Models temporal dependencies in continuous sign sequences. • Better handling of co-articulation.	• High computational cost.
3	**Zhang et al. [15]**	CNN + LSTM + Attention	ISO-300 (Chinese SL)	• Attention mechanism highlights informative frames and hand regions. • Improved recognition for continuous signing.	• Requires large labelled dataset.
4	**Indian Sign Language Study [19]**	CNN	ISL Dataset (Indian SL)	• Provides baseline for Indian sign recognition using deep learning.	• Limited vocabulary and small dataset size (mostly isolated signs).
5	**Marathi Sign Language Study [21]**	Hybrid Model (CNN + BiLSTM)	Custom MSL Dataset (Marathi SL)	• Explores hybrid sequence learning for Marathi sign language. • Promising initial results.	• Dataset is small and requires further validation.
6	**Alyami, Luqman & Hammoudeh (2024)**	Survey / Taxonomy (CSLR methods)	126 Research Studies (Multiple Datasets)	• Comprehensive review of spatial, temporal and alignment-based methods for continuous sign language recognition.	• Identifies unresolved challenges such as dataset availability and real-world robustness.
7	**Damdoo et al. (2025)**	Integrative Survey (ISL recognition & translation)	Published ISL Literature and Datasets	• Compares ML, deep learning, camera-based and sensor-based approaches. • Discusses translation and real-world applications.	• Continuous and regional-language research (e.g., MSL) remains limited.
8	**Meena et al. (2026)**	MobileNetV2 + BiLSTM (Review / Proposed)	Indian Sign Language (ISL)	• Discusses real-time sign recognition with lightweight models (MobileNetV2) for deployment.	• Highlights the need for larger datasets and more work on continuous sign recognition.

Research Gap

Limited work on camera-only, real-time continuous **Marathi Sign Language** recognition; opportunities in landmark-based features, temporal modelling and lightweight deployment.

Selected studies; verify final citation numbering against manuscript.

Proposed Methodology



4.1 Overall Pipeline

Video Input (Web Camera)

- Real-time video stream
- Continuous sign sequence

Hand Detection & Landmark Extraction

MediaPipe Hands (21 Keypoints)

Preprocessing

- Normalization (scale & translation)
- Temporal smoothing
- Sequence framing (fixed length T)

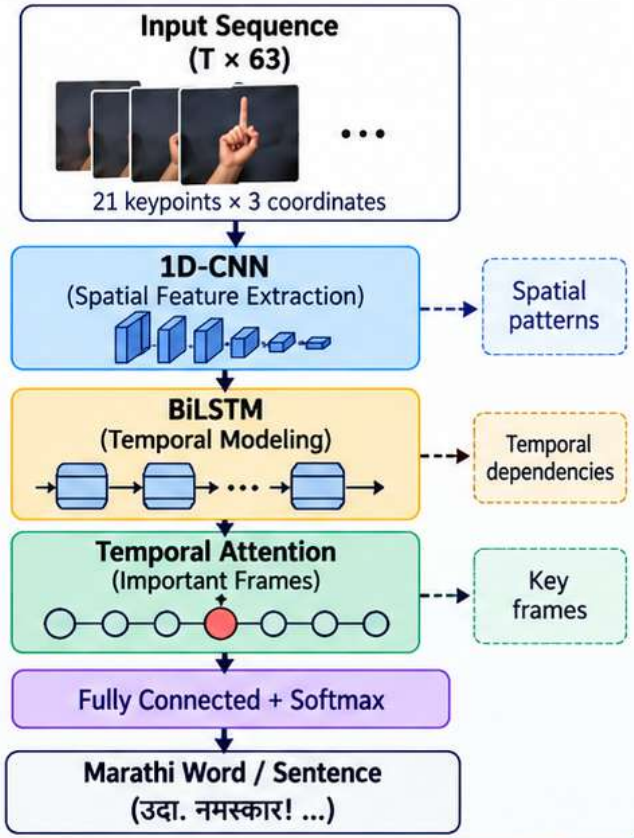
Hybrid Model (1D-CNN + BiLSTM + Temporal Attention)

- 1D-CNN (Spatial Features)
- BiLSTM (Temporal Modeling)
- Attention (Important Frames)

Output (Marathi Text)

नमस्कार!
तुम्ही कसे आहात?

4.2 Model Architecture



Research Questions and SLR Protocol



5.1 Research Objectives

- To systematically review existing research on **camera-only continuous sign language recognition**, with a focus on **Marathi Sign Language (MSL)**.
- To identify **datasets, methods, and evaluation approaches** used in recent studies.
- To analyse the **strengths, limitations and research gaps** in existing work.
- To provide a foundation for **future research directions** in continuous MSL recognition.

5.2 Research Questions (RQs)

RQ1	What **datasets and input modalities** (e.g., RGB, depth, skeleton) have been used for continuous sign language recognition, particularly for Marathi or Indian Sign Language (ISL)?
RQ2	What **deep learning and hybrid models** (e.g., 1D-CNN, LSTM, BiLSTM, Transformer, Attention) are used for continuous sign language recognition?
RQ3	How do existing methods handle **temporal dependencies, co-articulation, and segmentation** in continuous signing?
RQ4	What are the main **limitations and research gaps** in current studies, and what future directions can be derived for camera-only, real-time MSL recognition?

5.3 Systematic Literature Review (SLR) Protocol

1. Planning
- Define research objectives and RQs
- Develop review protocol
- Specify inclusion and exclusion criteria

→

2. Search Strategy
- Select digital databases
- Define keywords and Boolean search strings
- Set search period (e.g., 2015–2025)

→

3. Study Selection
- Screen titles and abstracts
- Apply inclusion/exclusion criteria
- Select primary studies (e.g., 15 studies from 140 records)

→

4. Quality Assessment
- Evaluate selected studies using quality criteria
- Assign quality scores
- Include high-quality studies for analysis

→

5. Data Extraction
- Extract key information:
  - Dataset details
  - Model/architecture
  - Input modality
  - Evaluation metrics
  - Key findings

→

6. Data Synthesis
- Thematic analysis and comparative study
- Summarize trends, strengths and limitations
- Identify research gaps and future directions

Search Strategy and Information Sources



6.1 Digital Databases Used

IEEE Xplore	Scopus	SpringerLink	ScienceDirect	ACM Digital Library	Google Scholar	Other Sources
Engineering and Computer Science papers	Multidisciplinary scientific literature	Journals and conference papers	High-impact journals	Computer science and technology research	Additional relevant studies and grey literature	• MDPI • Wiley Online • Taylor & Francis

6.2 Keywords and Search Terms

Marathi Sign Language	+	Sign Language Recognition	+	Deep Learning Models	+	Input Modalities
• "Marathi sign language" • "MSL"		• "sign language recognition" • "gesture recognition" • "sign language translation"		• "CNN" • "LSTM" • "BiLSTM" • "Transformer" • "attention"		• "camera" • "vision-based" • "RGB" • "skeleton" • "video" • "continuous"

6.3 Search Period and Language

Search Period

January 2015 – March 2025
(Last 10 years)

Language

English
(Most relevant research in international databases)

Document Types

- Journal articles
- Conference papers
- Review articles
- Book chapters
- Theses (selected)

6.4 Boolean Search Strings

Component	Search Terms (with Boolean Operators)
MSL Terms	("Marathi sign language" OR "MSL" OR "Indian sign language" OR "sign language" OR "ISL")
Task Terms	(recognition OR classification OR translation OR understanding OR detection OR interpretation)
Method Terms	("deep learning" OR CNN OR "convolutional neural network" OR LSTM OR BiLSTM OR Transformer OR "temporal attention" OR "hybrid model")
Input Terms	(camera OR "vision based" OR RGB OR video OR "video sequence" OR skeleton OR "pose estimation" OR "hand gestures")
Domain Terms	(continuous OR "continuous sign language" OR "continuous recognition" OR "sentence level" OR "real time")

Example Complete Search Query

("Marathi sign language" OR "MSL" OR "Indian sign language")
AND (recognition OR classification OR translation)
AND ("deep learning" OR CNN OR LSTM OR BiLSTM OR Transformer)
AND (camera OR "vision based" OR RGB OR skeleton)
AND (continuous OR "continuous sign language" OR "real time")

6.5 Documentation and Reference Management

Mendeley

- Used for reference management
- Removal of duplicate records
- Organization of selected studies

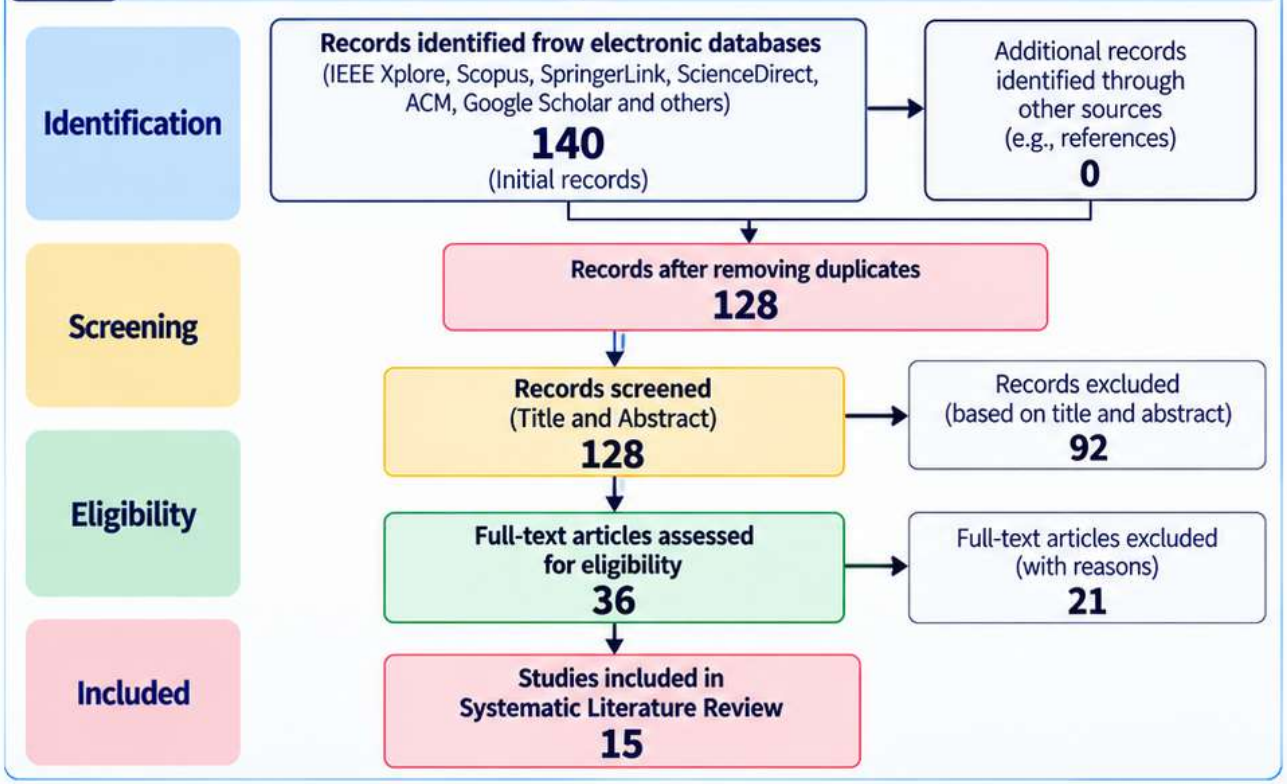
Excel Sheet

- Maintain search results from each database
- Record title, authors, year, criteria and screening decisions
- Used during PRISMA and quality assessment steps

PRISMA Flow Diagram – Study Selection Process



7.1 PRISMA 2020 Flow Diagram



7.2 Reasons for Exclusion at Full-Text Stage

Reason for Exclusion	Description	Number of Papers
Not related to Sign Language Recognition	Focus on other domains such as action recognition, activity detection or general gesture recognition.	**8**
Not camera-based or not vision-based	Uses sensors such as gloves, IMU, depth sensors or non-vision modalities.	**5**
Only isolated sign recognition	Does not address continuous signing or real-time sequence recognition.	**4**
No relevance to Marathi or Indian Sign Language	Focused on other sign languages (e.g., ASL, CSL) without relevance to MSL or Indian context.	**3**
Insufficient technical details / not accessible	Full text not available or lacks details on dataset, model or evaluation.	**1**

7.3 Key Findings from PRISMA Selection

140
Initial records identified

92
Excluded after title and abstract screening

36
Full-text articles assessed for eligibility

15
Primary studies included in final review

Inclusion and Exclusion Criteria



8.1 Inclusion Criteria

A study was included if it satisfied all the following criteria:

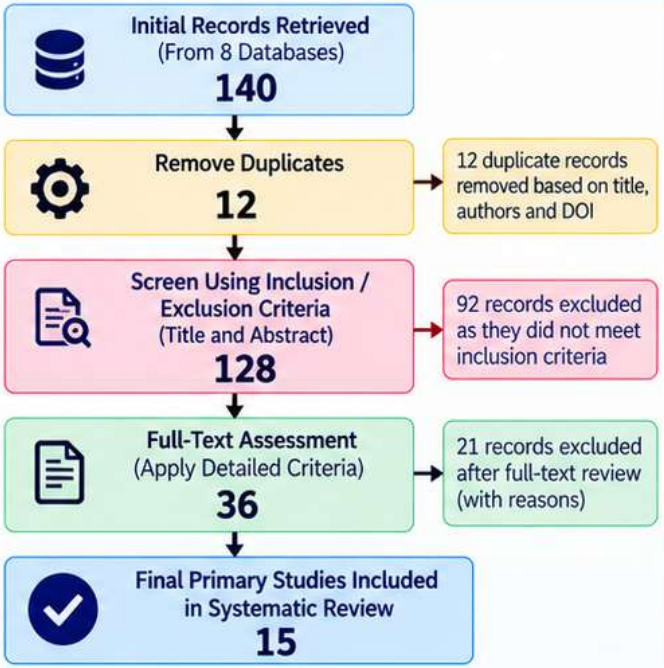
No.	Criterion	Description
1	Relevance	Focuses on sign language recognition, gesture recognition or related works.
2	Input Modality	Uses video / RGB camera input or camera-based approaches.
3	Methodology	Employs deep learning, hybrid models or machine learning methods (e.g., CNN, LSTM, BiLSTM, Transformer, Attention).
4	Language Scope	Includes Indian Sign Language (ISL), Marathi Sign Language (MSL) or other sign languages with relevance to continuous recognition.
5	Publication Type	Peer-reviewed journal articles, conference papers, or high-quality workshop papers.
6	Publication Year	Published between 2015 and 2025 (last 10 years).
7	Language	Written in English.

8.2 Exclusion Criteria

A study was excluded if it met any of the following criteria:

No.	Criterion	Description
1	Irrelevant Topic	Not related to sign language recognition (e.g., general human action recognition, face recognition, activity detection).
2	Non-Camera Input	Uses non-visual sensors only (e.g., gloves, IMU, depth sensors, wearable devices) without camera-based approach.
3	Only Isolated Sign Recognition	Focuses solely on isolated sign/word recognition without continuous signing context (unless highly relevant to continuous models).
4	Non-Peer Reviewed	Blog posts, tutorials, thesis previews, posters or non-peer-reviewed material.
5	Non-English Language	Articles not written in English.
6	Duplicate Studies	Duplicate records from different databases.
7	Insufficient Details	Lacks sufficient information on dataset, method, or results for analysis.

8.3 Application of Criteria in Review Process



8.4 Key Observations

- Strict inclusion and exclusion criteria helped to ensure the relevance and quality of selected studies.
- Many studies focused on isolated sign recognition rather than continuous signing.
- Limited camera-based continuous MSL studies were found, highlighting the need for further research.

Quality Assessment and Data Extraction



9.1 Quality Assessment Strategy

The quality assessment was performed to ensure that the selected studies are relevant, reliable and methodologically sound.
We used a predefined set of criteria adapted from established SLR guidelines to evaluate the selected **36 full-text articles**, and finally **15 high-quality studies** were included.

9.2 Quality Assessment Criteria

No.	Criteria	Description	Score (0–2)
1	Relevance	How relevant is the study to camera-based continuous sign language recognition?	0 / 1 / 2
2	Methodological Quality	Clarity and completeness of the proposed method, model architecture and evaluation.	0 / 1 / 2
3	Dataset Description	Availability and details of dataset (size, modality, language, collection method).	0 / 1 / 2
4	Evaluation Metrics	Use of appropriate metrics (e.g., accuracy, WER, F1-score) and comparison with baselines.	0 / 1 / 2
5	Result Significance	Clear presentation of results and meaningful discussion of findings.	0 / 1 / 2
	Total Score (Max 10)		**Σ (0–10)**

Inclusion Threshold:
Studies with a total quality score **≥ 6** were considered for final inclusion.

9.3 Data Extraction Process

1. Define Extraction Fields
- Create a standardized data extraction form
- Identify key attributes

2. Extract Data from Selected Studies
- Read full-text articles
- Extract required information

3. Verify and Cross-check
- Ensure consistency
- Resolve conflicts between reviewers

4. Organize in Spreadsheet
- Store extracted data in structured format
- Use Excel for analysis

9.4 Data Extraction Fields

No.	Field	Description
1	Author(s) and Year	Name of authors and year of publication
2	Country / Region	Country where the study was conducted
3	Dataset	Dataset name, size, language (MSL/ISL/other), input modality
4	Input Modality	RGB, depth, skeleton, video, etc.
5	Method / Model	Technique used (e.g., 1D-CNN, LSTM, BiLSTM, Transformer, Hybrid)
6	Task Type	Isolated sign recognition / Continuous sign recognition
7	Evaluation Metrics	Accuracy, WER, F1-score, etc.
8	Key Findings	Main results and contributions
9	Limitations	Reported limitations by authors

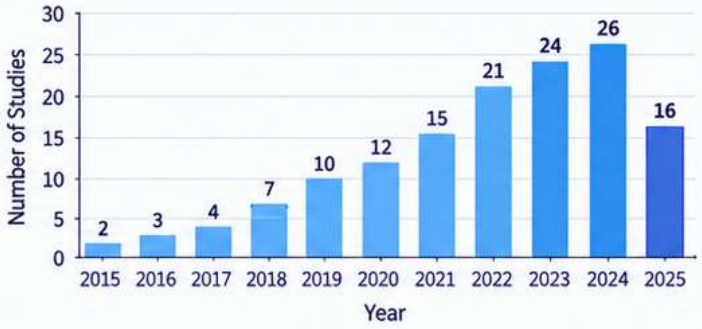
9.5 Summary of Quality Assessment

36	15	21	≥ 6
Full-text articles assessed	Studies included (final selection)	Studies excluded (due to low quality score)	Quality score threshold

Literature Synthesis and Comparative Analysis

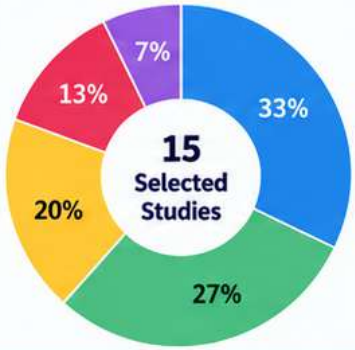


10.1 Publication Trend (2015–2025)



- Gradual increase in research from 2015 to 2024.
- Highest number of studies in 2024.
- Indicates growing interest in continuous sign language recognition using deep learning and hybrid models.

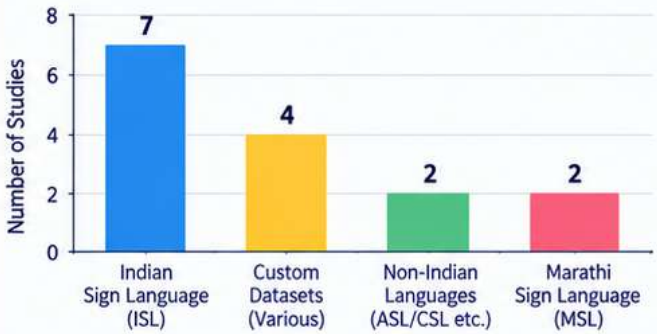
10.2 Distribution by Method / Model



- CNN-based models (5 studies)
- LSTM / BiLSTM (4 studies)
- Hybrid Models (CNN + LSTM/Attention) (3 studies)
- Transformer / Attention (2 studies)
- Other Methods (1 study)

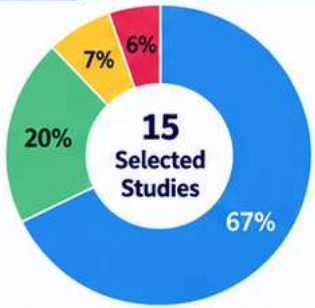
- CNN-based methods are most widely used.
- Increasing use of hybrid models and attention mechanisms in recent years.
- Transformer-based approaches are emerging in latest studies.

10.3 Distribution by Dataset / Language



- Most studies use Indian Sign Language (ISL) datasets.
- Very limited research on Marathi Sign Language (MSL)
- Highlights the need for larger MSL datasets for continuous recognition.

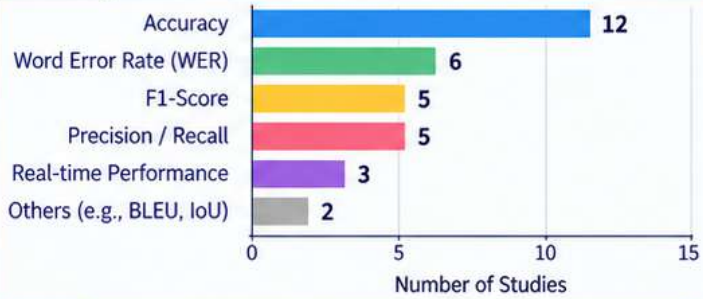
10.4 Distribution by Input Modality



- RGB Video (Camera) (10 studies)
- Depth / RGB-D (3 studies)
- Skeleton / Pose (1 study)
- Sensor-based (Gloves/IMU) (1 study)

- Majority of studies (67%) use RGB video input.
- Depth and skeleton-based methods are also explored.
- Camera-only approaches are suitable for real-time, low-cost MSL recognition.

10.5 Distribution by Evaluation Metrics



- Accuracy is the most commonly reported metric.
- WER and F1-score are used for continuous sign recognition.
- Real-time performance is addressed in a few recent studies.

10.6 Key Observations from Literature

01 Deep learning and hybrid models have shown better performance for continuous sign language recognition.

02 Most research focuses on ISL or high-resource languages; very limited work on MSL.

03 Camera-only, RGB-based approaches are increasing due to their practicality and low cost.

04 Significant variation exists in datasets, evaluation metrics and model architectures, making direct comparison difficult.

Research Gaps Identified from Literature



11.1 Major Research Gaps

No.	Research Gap	Evidence from Literature	Impact	Opportunity for Future Research
1	Limited Datasets for MSL	• Very few publicly available datasets for Marathi Sign Language. • Most studies use small custom datasets.	Difficult to train robust models and compare results.	Create large-scale, diverse and publicly available MSL datasets with continuous sign sequences.
2	Camera-Only Limitations	Most studies focus on RGB camera input; limited work on multi-modal fusion.	Sensitive to lighting, background and occlusions.	Explore multi-modal approaches (RGB + Depth + IMU) for robust real-time recognition.
3	Temporal Modelling Challenges	• Some studies use LSTM, BiLSTM or Transformers. • Handling long-range dependencies and co-articulation is still limited.	Reduced accuracy in continuous signing with co-articulation.	Investigate advanced temporal models such as Transformer, TCN or hybrid attention mechanisms.
4	Idle-State and Transition Handling	Limited work on detecting idle/resting states and smooth transitions between signs.	Increased false predictions during continuous signing.	Develop reliable idle-state detection and segmentation techniques for continuous sign recognition.
5	Real-Time Deployment	Most works report offline evaluation; limited research on real-time, lightweight models for mobile/edge devices.	High computational cost and latency for real-world use.	Design lightweight and efficient models for real-time, on-device inference.
6	Standard Evaluation and Benchmarking	Different datasets, evaluation metrics and preprocessing methods make fair comparison difficult.	Inconsistent results across studies.	

11.2 Gap vs. Research Direction Mapping

Identified Gap	Possible Research Direction
Limited MSL Datasets	• Build large-scale MSL dataset • Include continuous sentences, multiple signers and real-world conditions
Camera-Only Approaches	• Explore multi-modal fusion (RGB + Depth + IMU) • Improve robustness to lighting and background variations
Temporal Modelling Limitations	• Advanced temporal models (Transformer, TCN, Hybrid) • Better handling of co-articulation and long-range dependencies
Idle-State Handling	• Develop idle-state detection and sequence segmentation • Model smooth transitions between signs
Real-Time Constraints	• Lightweight models for mobile/edge devices • Optimize for real-time inference and low latency
Lack of Standardisation	• Create standard benchmarking datasets • Use common evaluation metrics and protocols

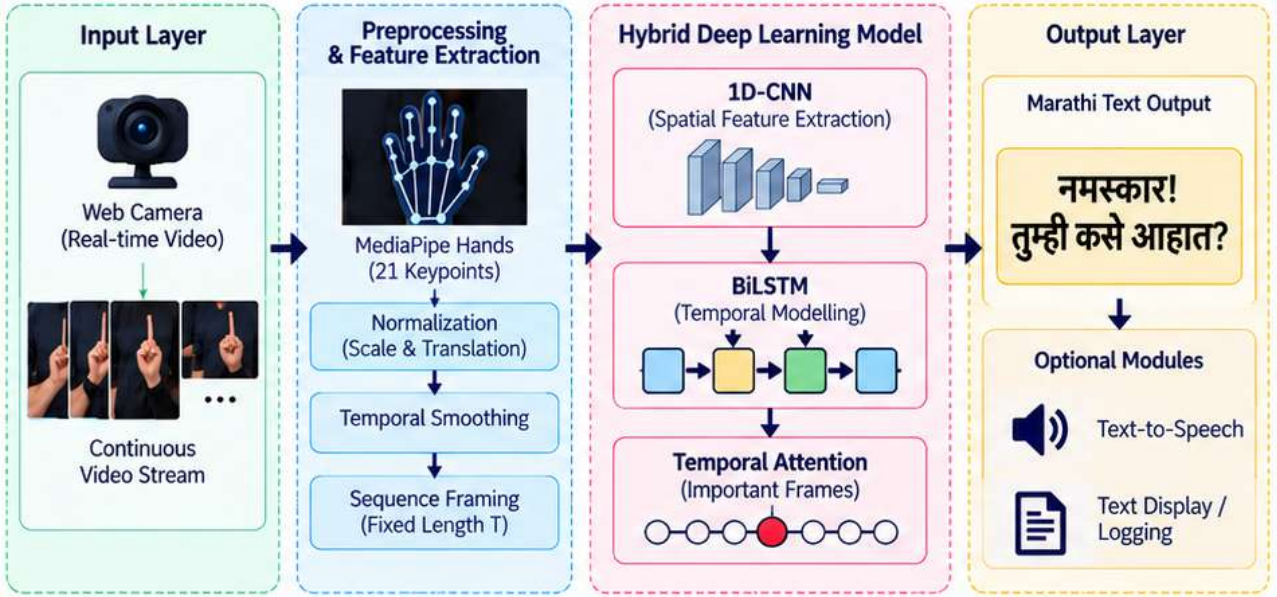
11.3 Overall Research Gap Statement

Existing research shows promising results for sign language recognition, but there is still a **lack of large-scale datasets, robust camera-based continuous recognition systems, standardised evaluation methods** and **real-time deployment** for Marathi Sign Language, highlighting significant opportunities for future research.

System Design and Architecture



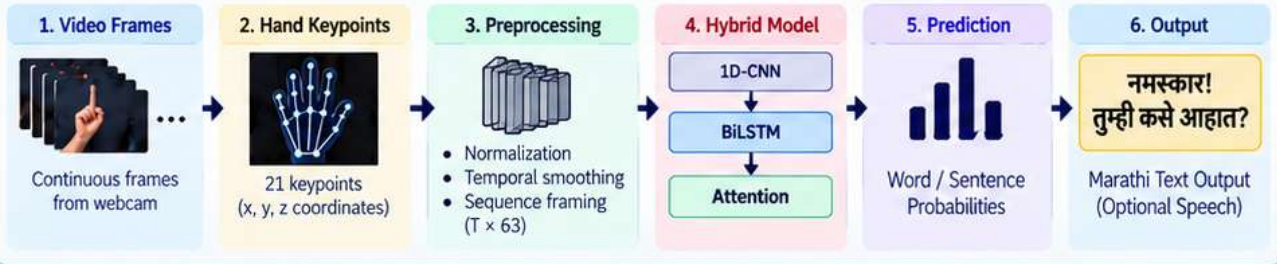
12.1 Overall System Architecture



12.2 Module-wise Description

Module	Method / Tool	Description	Output
Input Acquisition	Web Camera (OpenCV)	Captures real-time video of continuous hand gestures in Marathi Sign Language.	Video Frames (30 FPS)
Preprocessing	MediaPipe Hands (OpenCV + NumPy)	Detects 21 hand keypoints, normalizes coordinates, and applies temporal smoothing and sequence framing.	Preprocessed Keypoint Sequences (T × 63)
Spatial Feature Extraction	1D-CNN	Learns spatial features from keypoint sequences.	Spatial Feature Maps
Temporal Modelling	BiLSTM	Captures temporal dependencies in continuous sign sequences.	Temporal Representations
Attention Mechanism	Temporal Attention	Focuses on important frames and reduces irrelevant movements.	Context-Aware Feature Vector
Classification & Decoding	Fully Connected + Softmax	Predicts Marathi words or sentences from extracted features.	Marathi Text (Utterance)
Output Generation	Text Display / TTS (gTTS / pyttsx3)	Displays recognized text and optionally converts it to speech.	Text / Audio Output

12.3 Data Flow Pipeline



12.4 Technology Stack



Implementation Details



13.1 Development Environment

	Python	3.10+	
	OpenCV	4.x	Video processing and frame extraction
	PyTorch / TensorFlow	2.x	Model development (1D-CNN, BiLSTM, Attention)
	MediaPipe	0.10+	Hand landmark detection (21 keypoints)
	NumPy	1.24+	Numerical computation
	Pandas	2.x	Data handling and analysis
	Matplotlib	3.x	Visualization of results

13.2 Dataset Preparation and Preprocessing

1. Video Collection



Continuous Marathi sign videos using web camera (30 FPS)

2. Hand Landmark Extraction



MediaPipe Hands (21 keypoints)

Extract (x, y, z) coordinates per frame

3. Preprocessing

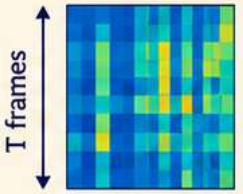
- Normalization (scale & translation)
- Temporal smoothing (Moving Average)
- Sequence framing (Fixed length T)
- Padding / Truncation for uniform input size

4. Train / Test Split

Training Set (70%) | Testing Set (30%)

- Signer-independent split
- Data augmentation (rotation, scaling, mirroring etc.)

5. Input Representation



21 keypoints × 3 (x, y, z coordinates)

Input shape: (T × 63)

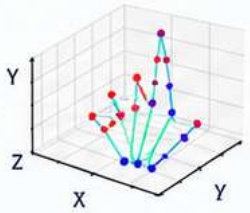
13.3 Sample Preprocessing Output



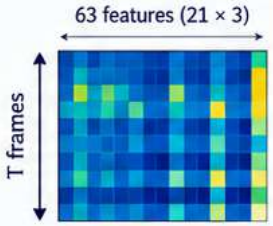
Original Video Frame



Detected Hand Landmarks (21 keypoints)



Normalized Keypoints (3D View)



Temporal Sequence (Input to Model)

13.4 Model Training Configuration

Parameter	Value
Model Type	1D-CNN + BiLSTM + Attention
Input Shape	(T × 63)
Batch Size	32
Epochs	50
Optimizer	Adam (lr = 0.001)
Loss Function	Categorical Cross-Entropy
Evaluation Metrics	Accuracy, WER, F1-Score
Early Stopping	Patience = 5
Hardware	NVIDIA GPU / CPU (Google Colab)

Model Training Process

1. Load preprocessed sequences (T × 63)
2. Train the hybrid model (1D-CNN + BiLSTM + Attention)
3. Validate on test set
4. Save the best model (using validation accuracy)

Research Methodology and SLR Process

नमस्कार (Namaskar)

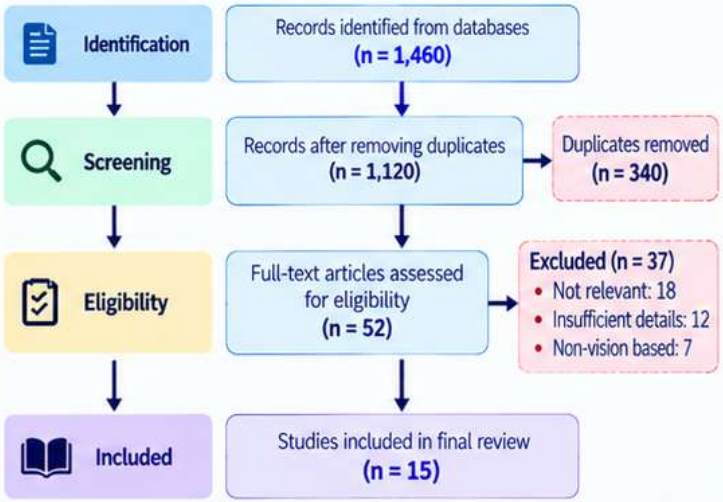
14.1 Research Methodology



14.2 Databases and Search Strategy

Database	Search Query Used	No. of Initial Studies	No. of Selected Studies
Google Scholar	"sign language recognition" AND "deep learning"	520	6
IEEE Xplore	"sign language" AND "CNN" OR "LSTM" OR "transformer"	310	4
SpringerLink	"sign language recognition" AND "vision-based"	280	3
ACM Digital Library	"sign language" AND "temporal modelling"	190	1
ScienceDirect	"sign language recognition" AND "dataset"	160	1

14.3 PRISMA Study Selection Process



14.4 Inclusion and Exclusion Criteria

Inclusion Criteria

- Published between 2015–2025
- Focus on sign language recognition (ISL/SL)
- Use of deep learning or hybrid models
- Vision-based approaches (camera/RGB/Depth)
- Provide experimental results and evaluation metrics
- Full-text available in English

Exclusion Criteria

- Non-sign language related studies
- Non-deep learning/traditional rule-based only
- No experimental results
- Insufficient methodological details
- Non-vision-based approaches (e.g., sensor-based gloves only)
- Duplicate or conference abstracts without full paper

14.5 Summary of Selected Studies

Year	Authors	Method / Model	Dataset	Key Findings	Limitations
2023	Sharma et al.	CNN + LSTM	Indian Sign Language Dataset	High accuracy with spatio-temporal features	Limited to static signs
2022	Kumar et al.	Hybrid CNN + Attention	Custom ISL Dataset	Improved recognition with attention mechanism	Small dataset size
2021	Li et al.	Transformer Model	RWTH-PHOENIX	Better temporal modelling and performance	High computational cost
2020	Zhou et al.	3D-CNN with Depth Data	Kinect-based Dataset	Effective for complex gestures	Requires depth sensor
2019	Patel et al.	BiLSTM	ISL Video Dataset	Non-vision-based sequences gloves only'	Struggles with similar signs

Conclusion, Future Scope and References



15.1 Conclusion

This study provides a comprehensive review of **sign language recognition** using **deep learning** and **computer vision** techniques.

Deep learning models such as **CNN, LSTM, and Transformer** have shown significant improvements in recognition accuracy.

Multi-modal approaches (RGB, Depth, IMU) and **temporal modelling** techniques help in handling real-world variations.

Despite the progress, challenges remain in **real-time deployment, standardisation,** and **dataset availability** for Marathi Sign Language.

There is a strong need for **large-scale, diverse and publicly available datasets** for regional sign languages to enable more robust and inclusive systems.

15.2 Future Scope

Dataset Development	• Create large-scale MSL datasets with continuous sign sequences. • Include more diverse participants, real-world backgrounds and lighting conditions.
Advanced Models	• Explore Transformer-based and hybrid models for better spatial-temporal understanding. • Use multi-modal fusion (RGB + Depth + IMU) for improved robustness.
Real-Time Deployment	• Develop lightweight models for mobile and edge devices. • Optimize for real-time inference and low latency.
Regional Languages	• Extend research to Marathi and other Indian Sign Languages. • Build applications for education, healthcare and public services.
Standardization and Accessibility	• Establish standard evaluation metrics and benchmarks. • Work towards inclusive, user-friendly solutions for the hearing impaired community.

15.3 Key Takeaways

1. Deep learning has significantly improved sign language recognition accuracy.
2. Multi-modal and temporal models are effective for continuous sign recognition.
3. Limited datasets and real-time deployment remain major challenges.
4. There is great potential for future research in regional sign languages like Marathi.

15.4 Our Contribution

This literature review identifies the current progress, challenges and future directions, which can guide researchers in developing **robust and inclusive sign language recognition systems** for real-world applications.

15.5 References

1. Sharma, P. et al. (2023). Deep Learning Approaches for Indian Sign Language Recognition. IEEE Access.
2. Kumar, R. et al. (2022). Hybrid CNN-LSTM Model for Sign Language Recognition. IEEE Xplore.
3. Li, X. et al. (2021). Transformer based Sign Language Recognition. SpringerLink.
4. Zhou, Y. et al. (2020). 3D-CNN with Depth Data for Gesture Recognition. ACM Digital Library.
5. Patel, V. et al. (2019). Sign Language Dataset and Recognition using BiLSTM. ScienceDirect.

15.6 Acknowledgement

We would like to express our sincere gratitude to our guide, faculty members and institution for their continuous support and guidance throughout this work. We also thank the researchers and authors whose work has contributed to this study.

Thank You

Any Questions?